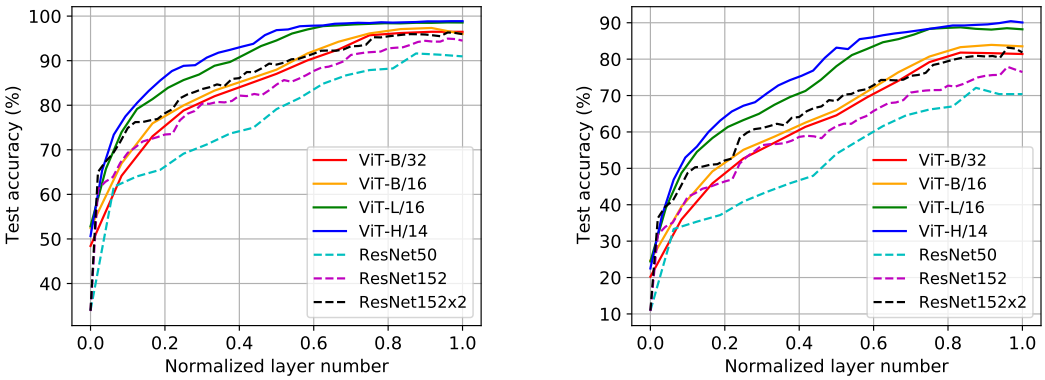
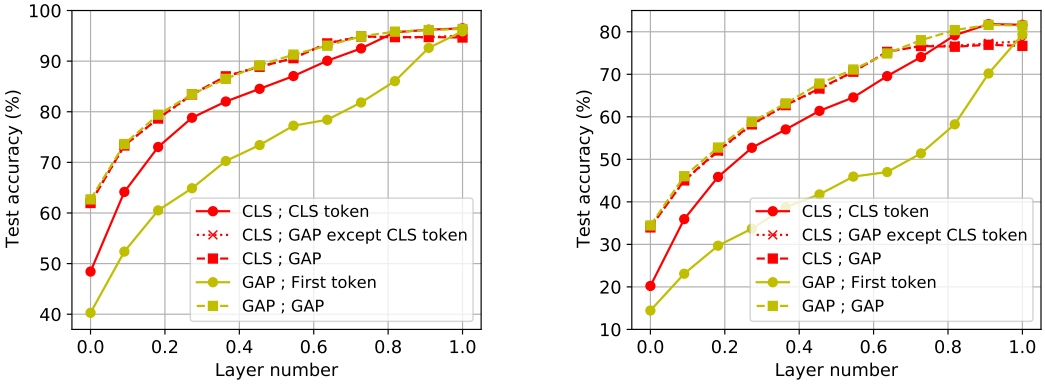


(a) CIFAR-10 (b) CIFAR-100

Figure F.2: Experiments with linear probes. Additional results on comparison of ViT and ResNet models (Fig. 13 right) on CIFAR-10/100 datasets.



(a) CIFAR-10 (b) CIFAR-100

Figure F.3: Experiments with linear probes. Additional results on comparison of ViT and ResNet models (Fig. 11 right) on CIFAR-10/100 datasets.

G Effects of Scale on Transfer Learning

Finally, we study how much representations change through the finetuning process for a model pre-trained on JFT-300M, finding significant variation depending on the dataset. For tasks like ImageNet or Cifar100 which are very similar to the natural images setting of JFT300M, the representation does not change too much. For medical data (Diabethic Retinopathy detection) or satellite data (RESISC45), the changes are more pronounced. In all cases, it seems like the first four to five layers remain very well preserved, even accross model sizes. This indicates that the features learned there are likely to be fairly task agnostic, as seen in Figure G.1 and Figure G.2.

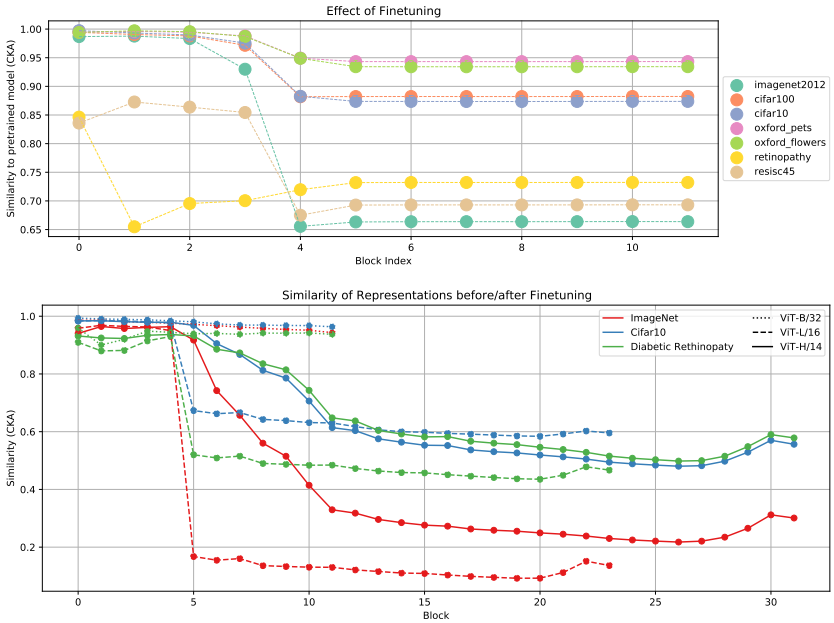


Figure G.1: (top) Similarity of representations at each block for ViT-B/16 models compared to before finetuning. (bottom) Similarity of representations at each block for different ViT model sizes.

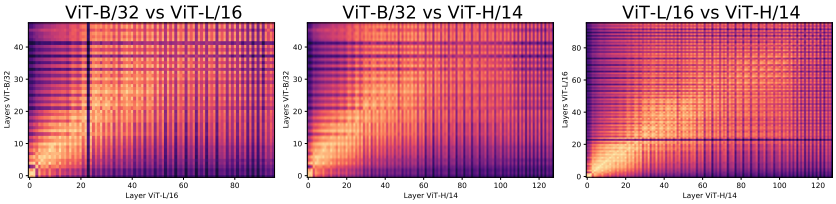


Figure G.2: Similarity of representations of different ViT model sizes.

H Preliminary Results on MLP-Mixer

Figure H.1 shows the representations from various MLP-Mixer models. The representations seem to also fall very clearly into distinct, uncorrelated blocks, with a smaller block in the beginning and a larger block afterwards. This is independent of model size. Comparing these models with ViT or ResNet as in Figure H.2 models makes it clear that overall, the models behave more similar to ViT than ResNets (c.f. Fig. 1 and 2).

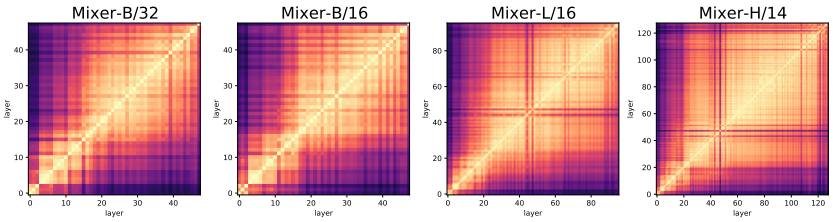


Figure H.1: Similarity of representations of different ViT model sizes.

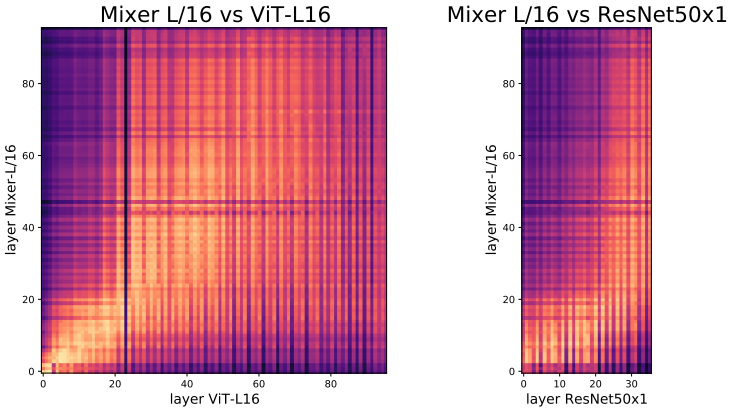


Figure H.2: Similarity of representations of different ViT model sizes.